Editor

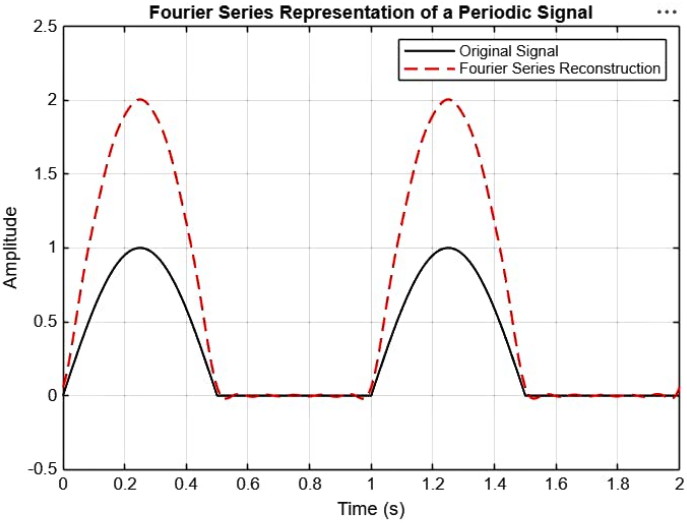
exp2_amplitude.m | exp2_add.m | exp_3.m | exp_4.m | exp_5_1.m | exp_5_2.m

/MATLAB Drive/exp_5_1.m

```
clc; clear; close all;
%% 5(i) Analysis of Fourier Series Representation of Continuous-Time
figure;
% Fourier Series Representation of a Periodic Signal
A = 1;
f0 = 1;
T = 1/f0;
t = linspace(0, 2*T, 2000);
% Half-wave rectified sinusoidal signal
x = A*sin(2*pi*f0*t);
x(x < 0) = 0;
N = 10;
a0 = (2/T)*trapz(t, x)/2;
an = zeros(1,N);
bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t, x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t, x.*sin(2*pi*k*f0*t));
end
xFS = a0*ones(size(t));
for k = 1:N
    xFS = xFS + an(k)*cos(2*pi*k*f0*t) ...
        + bn(k)*sin(2*pi*k*f0*t);
```

Figures

Figure 1



Command Window

Editor

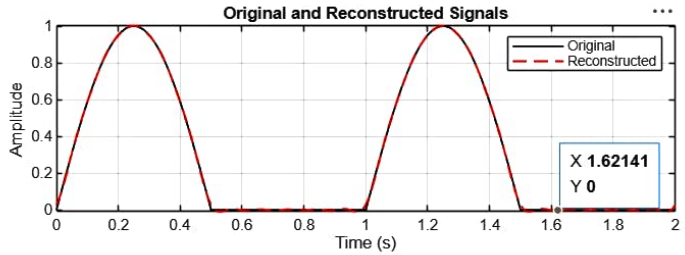
exp2_add.m | exp_3.m | exp_4.m | exp_5_1.m | exp_5_2.m | exp6_1.m

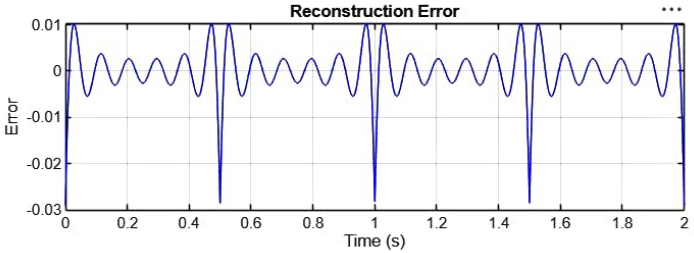
/MATLAB Drive/exp_5_2.m

```
clc; clear; close all;
%% 5(iv) Verification of Fourier Series Reconstruction and Analysis
figure;
% Fourier Series Reconstruction and Error Analysis
f0 = 1;
T = 1/f0;
t = linspace(0,2*T,4000);
x = max(sin(2*pi*f0*t),0);
N = 10;
a0 = (2/T)*trapz(t(1:2000),x(1:2000));
an = zeros(1,N); bn = zeros(1,N);
tt = t(1:2000); xx = x(1:2000);
for k = 1:N
    an(k) = (2/T)*trapz(tt,xx.*cos(2*pi*k*f0*tt));
    bn(k) = (2/T)*trapz(tt,xx.*sin(2*pi*k*f0*tt));
end
xr = a0/2*ones(size(t));
for k = 1:N
    xr = xr + an(k)*cos(2*pi*k*f0*t) + bn(k)*sin(2*pi*k*f0*t);
end
err = x-xr;
subplot(2,1,1);
plot(t,x,'k',t,xr,'r--','LineWidth',1.1);
```

Figures

Figure 1





Editor

exp2_add.m | exp_3.m | exp_4.m | exp_5_1.m | exp_5_2.m | exp6_1.m | unti

/MATLAB Drive/exp6_1.m

```
clc;
clear;
close all;

%% 6(i) Verification of Parseval's Relation for Periodic Signals

figure;

% Verification of Parseval's Relation
f0 = 1;
T = 1/f0;
t = linspace(0,T,4000);

x = max(sin(2*pi*f0*t),0);

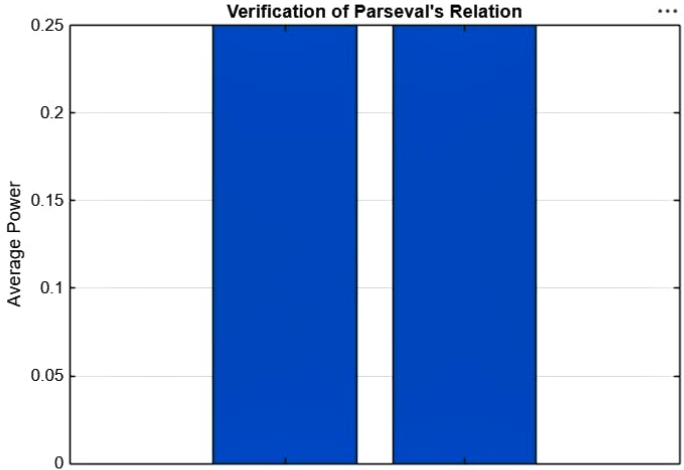
% Time-domain average power
P_time = (1/T)*trapz(t,abs(x).^2);

% Fourier-series coefficients
N = 20;

a0 = (2/T)*trapz(t,x);
```

Figures

Figure 1



Command Window

Editor

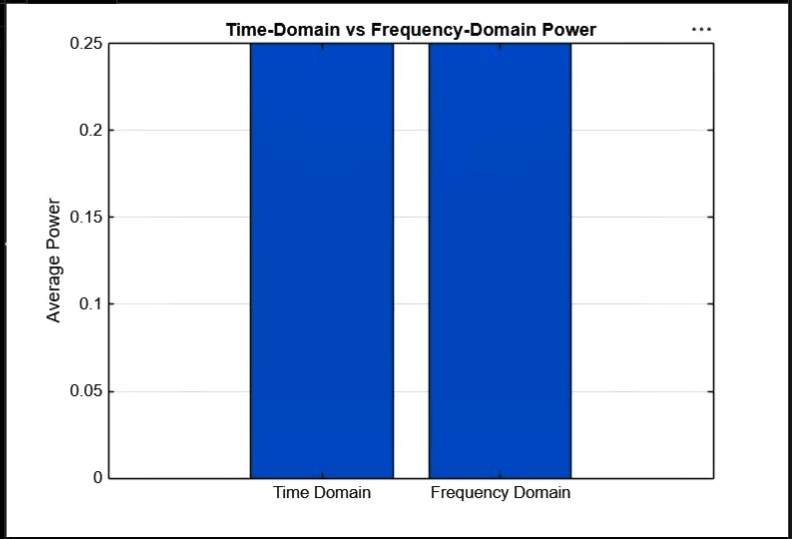
_3.m | exp_4.m | exp_5_1.m | exp_5_2.m | exp6_1.m | untitled | exp6_2.m

/MATLAB Drive/exp6_2.m

```
clc; clear; close all;
%% 6(iv) Comparison of Time-Domain Power with Frequency-Domain Harmo
figure;
% Time-Domain and Frequency-Domain Power Comparison
f0 = 1; T = 1/f0;
t = linspace(0,T,4000);
x = max(sin(2*pi*f0*t),0);
P_time = (1/T)*trapz(t,x.^2);
N = 20;
a0 = (2/T)*trapz(t,x);
an = zeros(1,N); bn = zeros(1,N);
for k = 1:N
    an(k) = (2/T)*trapz(t,x.*cos(2*pi*k*f0*t));
    bn(k) = (2/T)*trapz(t,x.*sin(2*pi*k*f0*t));
end
P_dc = a0^2/4;
P_harmonics = 0.5*sum(an.^2+bn.^2);
P_frequency = P_dc + P_harmonics;
fprintf('Time-domain power      = %.6f\n',P_time);
fprintf('DC power               = %.6f\n',P_dc);
fprintf('Harmonic power         = %.6f\n',P_harmonics);
fprintf('Frequency-domain power = %.6f\n',P_frequency);
bar([P_time P_frequency]);
```

Figures

Figure 1



Command Window

Editor

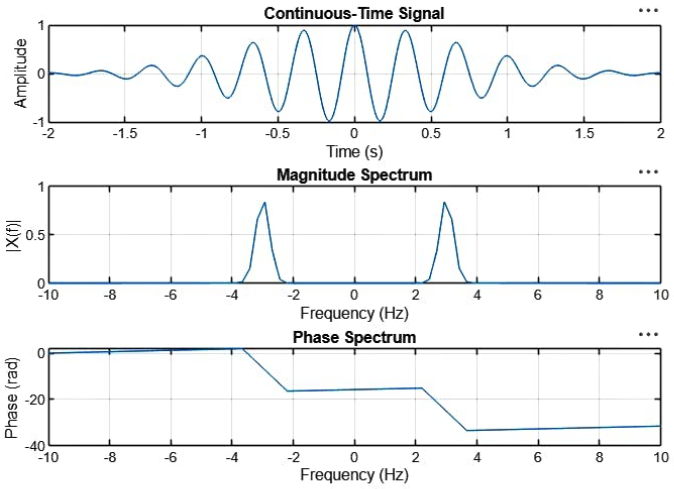
4.m | exp_5_1.m | exp_5_2.m | exp6_1.m | untitled | exp6_2.m | exp7_2.m

/MATLAB Drive/exp7_2.m

```
clc; clear; close all;
%% 7(ii) Determination of Magnitude and Phase Spectra of Continuous-
figure;
% Magnitude and Phase Spectrum of a CT Signal
t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2).*cos(2*pi*3*t);
N = 4096;
X = fftshift(fft(x,N))*dt;
f = (-N/2:N/2-1)/(N*dt);
magX = abs(X);
phaseX = unwrap(angle(X));
subplot(3,1,1);
plot(t,x); grid on;
xlabel('Time (s)'); ylabel('Amplitude');
title('Continuous-Time Signal');
subplot(3,1,2);
plot(f,magX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('|X(f)|');
title('Magnitude Spectrum');
subplot(3,1,3);
plot(f,phaseX); xlim([-10 10]); grid on;
xlabel('Frequency (Hz)'); ylabel('Phase (rad)');
```

Figures

Figure 1



Command Window

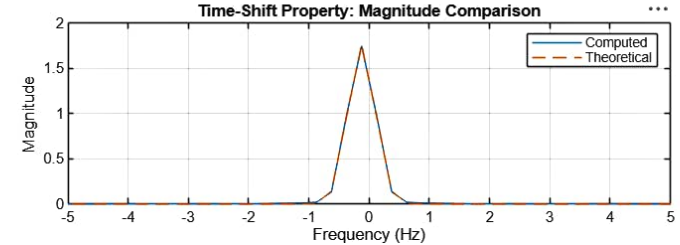
Editor

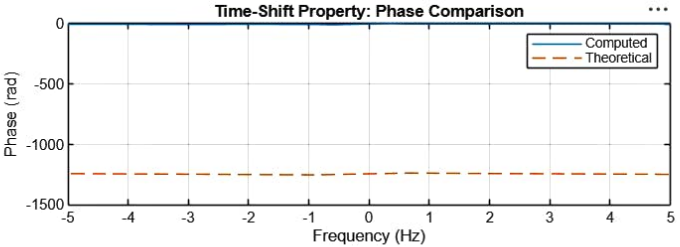
/MATLAB Drive/exp7_3.m

```
clc; clear; close all;
%% 7(iii ) Verification of Fourier Transform Properties and Frequenc
figure;
% Verification of Fourier Transform Properties
t = -2:0.001:2;
dt = t(2)-t(1);
x = exp(-t.^2);
x_shift = interp1(t,x,t-0.4,'linear',0);
X = fftshift(fft(x))*dt;
N = length(x);
f = (-N/2:N/2-1)/(N*dt);
% Time-shift property: x(t-t0) <-> X(f)e^(-j2*pi*f*t0)
X_shift_theory = X.*exp(-1j*2*pi*f*0.4);
X_shift = fftshift(fft(x_shift))*dt;
subplot(2,1,1);
plot(f,abs(X_shift),f,abs(X_shift_theory),'--');
xlim([-5 5]); grid on;
xlabel('Frequency (Hz)'); ylabel('Magnitude');
title('Time-Shift Property: Magnitude Comparison');
legend('Computed','Theoretical');
subplot(2,1,2);
plot(f,unwrap(angle(X_shift)),f,unwrap(angle(X_shift_theory)),'--');
xlim([-5 5]); grid on;
```

Figures

Figure 1





Command Window

Editor

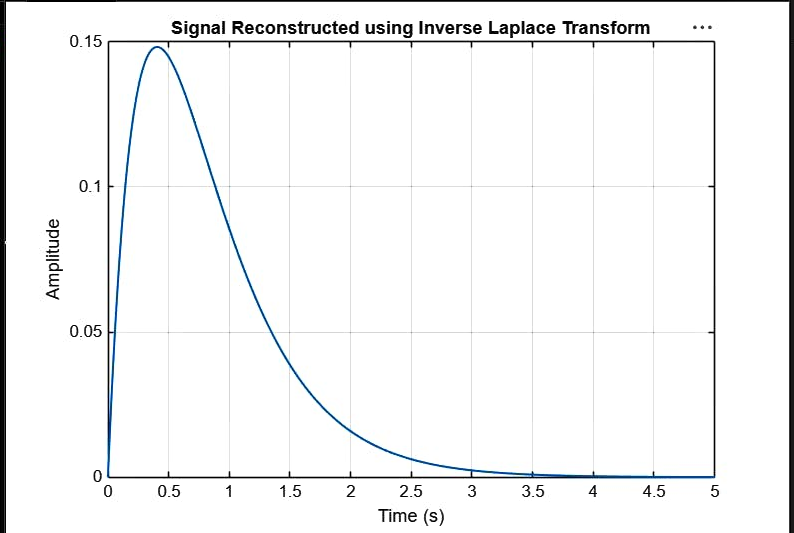
5_2.m | exp6_1.m | untitled | exp6_2.m | exp7_2.m | exp7_3.m | exp8_2.m

/MATLAB Drive/exp8_2.m

```
clc; clear; close all;
%% 8(ii) Determination of Inverse Laplace Transform for Signal Recon
figure;
% Inverse Laplace Transform
% Requires Symbolic Math Toolbox
syms s t
X = 1/((s+2)*(s+3));
x = ilaplace(X,s,t);
disp('Inverse Laplace Transform x(t) =');
pretty(x);
tt = 0:0.001:5;
xx = double(subs(x,t,tt));
plot(tt,xx,'LineWidth',1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');
```

Figures

Figure 1



Command Window

Editor

8_1.m | untitled | exp6_2.m | exp7_2.m | exp7_3.m | exp8_2.m | exp8_3.m

/MATLAB Drive/exp8_3.m

```
clc;
clear;
close all;

%% 8(iii) Analysis of Region of Convergence (ROC) and Its Significan

figure;

% X(s) = 1/((s+2)(s+4))
% Poles are at s = -2 and s = -4

p = [-4 -2];

plot(real(p), imag(p), 'bx', 'MarkerSize', 10, 'LineWidth', 2);
hold on;

xline(-4, '--b', 'LineWidth', 1.5);
xline(-2, '--b', 'LineWidth', 1.5);

grid on;
box on;

xlabel('Real(s)');
```

Figures

Figure 1

